

Self-Contrastive Learning to Boost Weakly Supervised Anomaly Detection

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Abstract—Weakly supervised anomaly detection methods (WADMs) can effectively utilize incomplete label data to address the issue of imbalanced samples, thereby reducing the reliance on the quantity of labeled data and demonstrating superior anomaly detection capabilities in network and service management. However, when confronted with significant label noise and missing labels, existing WADMs still struggle to adequately extract the deep feature information of samples, leading to a decline in model performance. To tackle this challenge, we propose a self-contrastive enhanced weakly supervised anomaly detection framework, called SEAD-Net, which enhances the feature representation capability of data in the model's feature space, thereby improving the accuracy and robustness of anomaly detection. Specifically, we first design a personalized data enhancement module that augments data representation by applying various transformations to the raw data. Subsequently, a self-contrastive enhanced learning module is introduced to impose hybrid constraints on the augmented samples, constructing the overall distribution structure while learning deep sample feature spaces under complex scenario disturbances. Finally, we extract contrastive enhancement features within the deep sample feature space and perform probabilistic generation to enable effective decision-making via an anomaly probability generation module. Experimental results on a series of public benchmark datasets demonstrate that our SEAD-Net outperforms the second-best WADM by 5.95% in average AUC-ROC and 16.28% in average AUC-PR.

Index Terms—Anomaly detection, deep learning, data enhancement, self-contrastive learning.

I. INTRODUCTION

ANOMALY detection significantly enhances data quality and system reliability by enabling the timely

identification of potential anomalies, thus attracting extensive research. As the complexity and scale of data continue to grow, the application of advanced anomaly detection techniques becomes crucial for ensuring the effectiveness of network and service management systems [1], [2], [3]. This importance is further highlighted in various engineering applications, such as blockchain networks [4], mobile network management [5], the Internet of Things (IoT) [6], [7], large-scale services [8], and large-scale engineering collaboration projects [9]. In network and service management, anomaly detection plays a vital role in real-time monitoring and fault diagnosis, helping to prevent network performance degradation and service quality deterioration. Additionally, it enables real-time monitoring and analysis of Quality of Service (QoS) parameters, thereby ensuring the stability and efficiency of network services. As a result, researchers propose numerous anomaly detection methods, which not only enhance the performance of anomaly detection but also significantly improve the response speed to data anomalies, thereby reducing the costs associated with managing and handling anomalous behaviors [10].

Traditional anomaly detection methods primarily determine whether data deviates from a normal distribution by analyzing its statistical properties [11]. However, these methods exhibit significant disadvantages when handling large-scale, high-dimensional data. In contrast, Deep Learning (DL)-based anomaly detection methods can model nonlinear data, automatically learn deep features, and uncover hidden distributions in large-scale data, thereby improving anomaly detection performance [12]. Recent application-oriented studies further demonstrate the effectiveness of learning-based representations in related scenarios, such as encrypted industrial traffic anomaly detection [13] and indoor localization of uncrewed aerial vehicles [14]. Specifically, these data-driven methods can be categorized into three groups: unsupervised, supervised, and weakly supervised methods. Unsupervised methods identify anomalies by modeling normal samples and capturing the overall structure and latent representations of the data [15], [16], [17]. However, these methods have poor generalization capabilities due to their difficulty in capturing complex data relationships and their inability to utilize label information. Supervised techniques involve training models with labeled normal and anomalous samples, allowing them to adjust to diverse data distributions and improve overall performance [18], [19]. Nevertheless, these methods necessitate labels of high integrity and precision. In scenarios where labels are absent or corrupted, performance experiences a noticeable decline due to the dependency of supervised methods on

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accurate labels for maintaining correct associations between inputs and outputs [20].

To address challenges such as label scarcity, contamination, and class imbalance, researchers have turned to Weakly Supervised Anomaly Detection Methods (WADMs) [21], [22]. These approaches utilize Convolutional Neural Networks (CNNs) [23], Residual Networks (ResNets) [24], or Autoencoders (AEs) [25] to construct feature extractors that capture essential information from samples and feed it into anomaly detection classifiers. WADMs integrate the advantages of both unsupervised and supervised methods, alleviating the poor generalization capability of unsupervised methods while effectively utilizing incomplete label data to handle imbalanced samples. Nevertheless, the existing WADMs still exhibit the following limitations:

- Incomplete and inaccurate data annotation stems from the difficulty and high cost of obtaining data labels, which leads to a decrease in WADMs model performance.
- The scarcity of data annotation and insufficient model learning capabilities limit WADM's ability to fully extract the complete distribution and deep feature information of the data, adversely affecting its generalization performance and robustness.

Recently, the remarkable success of self-supervised pre-training demonstrates that instance-level contrastive learning can become a powerful feature extractor even without manual labels, thereby enhancing the ability to capture the complete data distribution and deep feature representations. Pioneering frameworks such as MoCo [16] and SimCLR [17] employ memory queues, momentum-updated encoders, and aggressive data augmentations to learn discriminative representations by contrasting positive and negative pairs. Although these methods are originally designed for large-scale image recognition, their core principles, namely dynamic dictionary maintenance and robust augmentation invariance, provide valuable insights for anomaly detection. Therefore, we migrate and adapt these mechanisms to the WADM scenario by customizing the augmentation pool to accommodate streaming data characteristics and incorporating the contrastive objective into the pretraining stage. An effective balancing coefficient trades off between preserving fine-grained details (reconstruction) and learning separable latent structures (contrast), thereby leading to performance improvements on benchmark anomaly detection datasets.

Consequently, we propose a Self-contrastive Enhanced Weakly Supervised Anomaly Detection framework, termed SEAD-Net. SEAD-Net extracts comprehensive data representations through a customized Data Enhancement Module (DEM) and a Self-contrastive Enhancement Learning Module (SCLM). Specifically, DEM is utilized to enrich and enhance the robustness of input data. By applying various transformations to the raw data, it introduces real-world noise and anomalies, effectively compensating for deficiencies in data feature representation. Furthermore, existing anomaly detection models typically emphasize learning distinctions among samples from different categories while overlooking the intrinsic relationships among samples within the same category. However, our SCLM ensures comprehensive learning of all

category structures and enables SEAD-Net to autonomously distinguish differences among samples by learning both surface and deep features. Importantly, these two modules are complementary: DEM provides additional learnable information to SCLM, while SCLM amplifies the enhancement benefits offered by DEM. Finally, an anomaly probability generation module processes the features to obtain anomaly probabilities, which guide the model's decision-making. The main contributions of this work are as follows:

- We propose a self-contrastive enhanced weakly supervised anomaly detection framework, called SEAD-Net, which enhances the feature representation capability of data in the model's feature space, thereby improving the accuracy and robustness of anomaly detection.
- We introduce a self-contrastive enhanced learning module for SEAD-Net, which utilizes hybrid constraints to enable the model to autonomously discern differences and similarities among normal data samples, encouraging deeper feature learning.
- We conduct extensive experiments and analyses, demonstrating that our SEAD-Net achieves superior detection performance compared to other methods.

The remainder of this paper is organized as follows: Section II reviews the related work on anomaly detection methods. Section III provides a detailed description of the proposed SEAD-Net. Sections IV and V analyze the experimental results of SEAD-Net and provide corresponding discussions, respectively. Finally, Section VI concludes the paper.

II. RELATED WORKS

A. Unsupervised Anomaly Detection Approaches

Unsupervised anomaly detection algorithms have garnered significant attention due to their advantages, including the lack of reliance on expert-annotated sample data, adaptability to changing environments, the ability to discover unknown anomalies, and the capacity to handle rare events. Li et al. [26] proposed COPOD, which models multivariate distributions using empirical Copulas and quantifies data extremity via tail probabilities for efficient detection. Li et al. [27] developed ECOD, which estimates anomaly scores through empirical cumulative distribution functions across dimensions. Liu et al. [28] introduced an unsupervised anomaly detection method, called "iForest", which explicitly isolates anomalous samples by constructing a collection of isolation trees. Xu et al. [29] extended this concept to DIForest, leveraging deep neural networks to map data into nonlinear feature spaces for improved isolation. Ahmed et al. [30] designed a graph-regularized autoencoder that enhances anomaly detection and clustering by preserving local neighborhood structures. Zong et al. [31] proposed DAGMM, an unsupervised anomaly detection technique that uses a deep AE to extract latent representations and reconstruction errors for each sample, optimizing the parameters of both the AE and a Gaussian mixture model in an end-to-end manner. Schlegl et al. [32] introduced AnoGAN, an unsupervised deep convolutional Generative Adversarial Network (GAN) with an anomaly scoring scheme that maps from

image space to latent space, effectively capturing local anomalies and improving the recognition rate of abnormal images. While unsupervised learning methods offer a cost-effective and efficient training approach, overlapping distributions between normal and anomalous samples, ambiguous class boundaries, and the presence of substantial noise or contamination in the observed data make reliable separation difficult.

B. Supervised Anomaly Detection Methods

These methods rely on labeled training data to identify anomalies by learning the differences between normal and abnormal samples. Compared with unsupervised methods, supervised methods can generally achieve better stability and performance when sufficient labels are available. For instance, Liu et al. [33] proposed a quasi-periodic time-series anomaly detection framework that combines an innovative segmentation algorithm with a hybrid attention model to achieve high accuracy. Steinwart et al. [34] reformulated anomaly detection as a binary classification problem and developed a support vector machine (SVM)-based approach with a sampling strategy for artificially labeled data. Sundarkumar et al. [35] developed a hybrid model combining k-reverse nearest neighbors and one-class SVM, and demonstrated high sensitivity in fraud-related prediction tasks. Yang et al. [36] introduced an active learning method for anomaly detection in the Internet of Things (IoT), which improves labeling efficiency under data imbalance. Ding et al. [37] proposed a multi-generator generative adversarial network (GAN) framework that integrates a classifier and a feature extractor, and improves inter-class separability for intrusion detection. Overall, supervised anomaly detection performs well when labeled data are sufficient, task objectives are clear, and sample classes are relatively balanced, but its applicability remains limited when anomaly labels are extremely scarce and expensive to obtain.

C. Weakly Supervised Anomaly Detection Methods

WADM is an anomaly detection paradigm that lies between supervised and unsupervised learning methods. It addresses the problem of data imbalance by leveraging a small amount of labeled data and a large amount of unlabeled data to train models. Starting with an illustration, Snell et al. [38] proposed a prototype network, FSNet, to address few-shot learning by classifying samples based on distances to learned class prototypes, thereby achieving effective generalization under limited supervision. Pang et al. [39] introduced REPEN, an efficient weakly supervised anomaly detection method that combines ranking-based representation learning with anomaly detection. By exploiting a small amount of labeled data as prior knowledge, REPEN learned more expressive and task-relevant representations. Building on this line of work, Pang et al. [40] developed DevNet, which leveraged a limited number of labeled anomalies and prior probabilities to perform end-to-end anomaly score learning through neural bias optimization, rather than focusing solely on feature extraction. Zhou et al. [41] proposed an anomaly detection network, FEWAE, which utilized a pre-trained model to pre-encode and extract features from data. It effectively transformed

latent information from deep autoencoders into crucial coding elements, thereby enhancing detection performance. Ruff et al. [42] proposed Deep-SAD, a deep semi-supervised anomaly detection method that integrated a small number of labeled anomalies and employed an information-theoretic loss to minimize the entropy of normal samples while maximizing that of anomalous ones, thus improving class separability. Li et al. [43] introduced Dual-MGAN, a GAN-based semi-supervised anomaly detection method that trained two generators to model the distribution of normal data for anomaly discrimination. It incorporated distribution construction and data augmentation modules to enhance weakly supervised learning across diverse datasets. Mirzaei et al. [44] utilized only normal-class labels and employed a text-image diffusion model to convert normal samples into “pseudo anomalies” that were semantically and pixel-wise close to real anomalies, enabling robust anomaly detection without access to real anomalous data.

WADMs can effectively reduce the cost of obtaining data labels and have a broader range of applications. However, current WADMs primarily focus on learning shallow feature representations and do not fundamentally address the issue of insufficient deep structural feature capture caused by the complexity of the data itself. Compared with existing methods, our proposed SEAD-Net first customizes specific data enhanced representations for different data sources, making complex data structures more amenable to be learned. Subsequently, a self-contrastive enhanced learning module is introduced to further explore the latent feature representations of the data, effectively addressing the issue of insufficient model depth in WADMs.

III. ARCHITECTURE OF SEAD-NET

A. Overview of SEAD-Net

As shown in Fig. 1, SEAD-Net includes the three key modules: a Data Enhancement Module (DEM), a Self-Contrastive Enhancement Learning Module (SCLM), and an Anomaly Probability Generation Module (APM). Specifically, the customized DEM applies several augmentations to generate the enhanced representation $\mathbf{x}'$ of the input data $\mathbf{x}$. Subsequently, $\mathbf{x}'$ passes through the SCLM to learn deep representations of the samples. Note that the SCLM, within a traditional deep autoencoder framework, combines reconstruction and contrastive constraints to transform the input sample into more meaningful enhanced representations $\mathbf{f}$. Finally, APM utilizes the obtained enhanced feature representation $\mathbf{f}$, along with other factors: the reconstruction error e and the reconstruction residuals vector ϵ , to calculate the anomaly probability y' of the original data $\mathbf{x}$, which enables effective decision-making on anomalous samples.

B. Data Enhancement Module

To address the issues of incomplete labels and noise contamination in the input data, we elaborately design a Data Enhancement Module (DEM) to improve sample diversity and robustness. Specifically, DEM applies a series of transformations and patterns to the original dataset T , including representative operations such as *Random Crop*, *Random*

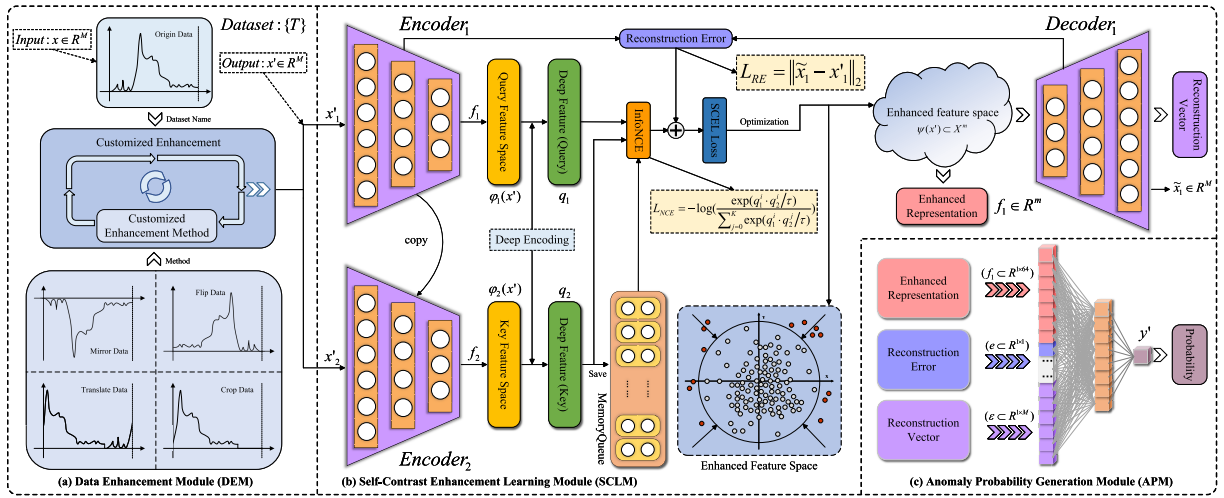


Fig. 1. The proposed SEAD-Net architecture consists of the customized Data Enhancement Module (DEM), the Self-Contrastive Enhancement Learning Module (SCLM), and the Anomaly Probability Generation Module (APM). Specifically, the DEM enhances the input data x , after which the SCLM extracts enhanced features f_1 , the reconstruction error e , and the reconstruction residuals vector ϵ from the augmented inputs x_1' and x_2' . Finally, the APM generates the anomaly probability y' based on the extracted features (f_1, e, ϵ).

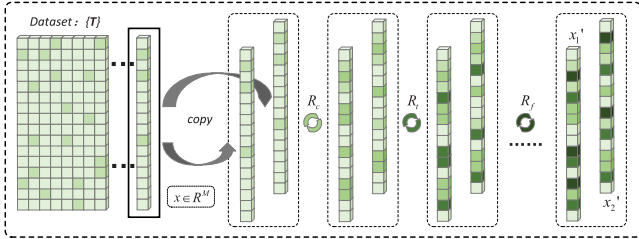


Fig. 2. An illustrative representation of DEM. The original input data $\mathbf{x}$ from the dataset $\mathbf{T}$ undergoes a series of carefully designed augmentation operators: R_c, R_t, R_f, M_i , generating a pair of augmented data $\mathbf{x}_1'$ and $\mathbf{x}_2'$.

Translation, Random Flip, and Mirroring, as well as additional transformations from the augmentation library such as Random ZeroReplace, Random Noise, Random Permuted, Random Contrast, Random Scale, Random Brightness, Fixed Scale, Opposite Time, and Random BlockAug, which better simulate the complexity of real-world data. To better illustrate the enhancement process, we select four representative transformations as examples. As shown in Fig. 2, DEM transforms the samples $\mathbf{x} \in \mathbb{R}^M$ from $\mathbf{T}$ into augmented data $\mathbf{x}' \in \mathbb{R}^M$ through a series of augmentation processes, where M represents the dimensionality of the sample. We denote this series of augmentations by the augmentation operator ∇_{aug} :

$$\mathbf{x}' = \nabla_{\text{aug}}(\mathbf{x}) = R_c \circ R_t \circ R_f \circ M_i(\mathbf{x}), \quad (1)$$

where, $\circ$ denotes the sequential application of these processes; and R_c, R_t, R_f, M_i are defined as:

$$R_c(\mathbf{x}) = \text{Random Cropping}(\mathbf{x}, \text{crop_dim}); \quad (2)$$

$$R_t(\mathbf{x}) = \text{Random Translation}(\mathbf{x}, \text{max_shift}); \quad (3)$$

$$R_f(\mathbf{x}) = \text{Random Flip}(\mathbf{x}, \text{flip_axis}); \quad (4)$$

$$M_i(\mathbf{x}) = \text{Mirroring}(\mathbf{x}, \text{mirror_axis}). \quad (5)$$

Additionally, the definitions of other augmentation methods in the enhancement library are as follows:

$$R_n(\mathbf{x}) = \text{Random Noise}(\mathbf{x}, \text{gaussian}, \text{noise_level}); \quad (6)$$

$$O_t(\mathbf{x}) = \text{Opposite Time}(\mathbf{x}, \text{time_axis}); \quad (7)$$

$$R_p(\mathbf{x}) = \text{Random Permuted}(\mathbf{x}, \text{block_size}); \quad (8)$$

$$R_z(\mathbf{x}) = \text{Random ZeroReplace}(\mathbf{x}, \text{replace_prob}); \quad (9)$$

$$R_{ct}(\mathbf{x}) = \text{Random Contrast}(\mathbf{x}, \text{contrast_range}); \quad (10)$$

$$R_b(\mathbf{x}) = \text{Random Brightness}(\mathbf{x}, \text{brightness_range}); \quad (11)$$

$$R_s(\mathbf{x}) = \text{Random Scale}(\mathbf{x}, \text{scale_factor_range}); \quad (12)$$

$$F_s(\mathbf{x}) = \text{Fixed Scale}(\mathbf{x}, \text{scale_factors}); \quad (13)$$

$$R_{ba}(\mathbf{x}) = \text{Random BlockAug}(\mathbf{x}, \text{num_slices}, \text{aug_methods}). \quad (14)$$

Here, *Random Crop* randomly selects a continuous segment of length *crop_dim* from the input data $\mathbf{x}$ and pads it with zeros to match the original sequence length; *Random Translation* cyclically shifts $\mathbf{x}$ along the temporal axis by a random offset *max_shift*; *Random Flip* flips the sequence $\mathbf{x}$ along the central axis *flip_axis*; *Mirroring* performs a reflection of $\mathbf{x}$ along the axis *mirror_axis*; *Random Noise* adds stochastic perturbations of type *gaussian* with intensity *noise_level* to $\mathbf{x}$; *Opposite Time* reverses $\mathbf{x}$ along the temporal axis *time_axis*; *Random Permuted* partitions $\mathbf{x}$ into fixed-size blocks *block_size* and randomly permutes the first $N-1$ blocks; *Random ZeroReplace* randomly replaces elements in $\mathbf{x}$ with zeros at a probability *replace_prob*; *Random Contrast* adjusts the contrast of $\mathbf{x}$ by a factor within *contrast_range*; *Random Brightness* modifies the brightness of $\mathbf{x}$ by adding a random offset within *brightness_range*; *Random Scale* rescales the amplitude of $\mathbf{x}$ using a random scaling factor sampled from *scale_factor_range*; *Fixed Scale* applies a fixed scaling factor randomly selected from a predefined set *scale_factors*; and *Random BlockAug* divides $\mathbf{x}$ into *num_slices* blocks, randomly rearranges the first $N-1$ blocks, and applies a specified set of augmentation methods *aug_methods* to each block.

For each original input data $\mathbf{x}$, DEM introduces the two parallel transformations:

$$\begin{cases} \mathbf{x}'_1 = \nabla_{\text{aug}}(\mathbf{x}); \\ \mathbf{x}'_2 = \nabla_{\text{aug}}(\mathbf{x}), \end{cases} \quad (15)$$

where $\mathbf{x}'_1$ and $\mathbf{x}'_2$ are two corresponding augmented data. Due to inherent randomness, the same augmentation steps applied to $\mathbf{x}$ twice may yield different results. It is important to note that we manually select different customized augmentation steps for each dataset based on specific requirements to ensure the effectiveness of the augmentation process. The necessity and effectiveness of these dataset-specific customization strategies are further verified in the ablation study of Section V-B and in Section S-V of the **Supplementary Material**.

C. Self-Contrastive Enhancement Learning Module

As illustrated in Fig. 1, the augmented data $\mathbf{x}'_1$ and $\mathbf{x}'_2$ undergo parallel dual-constraint branches for self-contrastive enhancement learning: 1) Reconstruction branch, which is used to construct the sample feature space (φ); 2) Contrastive branch, which is used to enhance the sample feature space (ψ). Here, the enhanced feature space, generated through multi-constraint training, represents the intrinsic structure and correlations of the input data.

1) *Reconstruction Branch*: Within the autoencoder $\mathcal{A}_e$, samples are constructed into a more compact latent feature space. Specifically, $\mathcal{A}_e$ consists of an Encoder and a Decoder:

- **Encoder**: The encoder is responsible for mapping the input sample $\mathbf{x}'_i \in \mathbb{R}^M$ to a lower-dimensional feature space $\varphi(\mathbf{x}')$, extracting the compressed representation $\mathbf{f}_i \in \mathbb{R}^m$. The encoding process is defined as $g_e(\cdot; \mathbf{W}_e) : X^M \rightarrow \varphi(\mathbf{x}')$, where X^M is the original sample space of $\mathbf{x}$, with M denoting the dimensionality of $\mathbf{x}$ and m denoting the dimensionality of $\mathbf{f}_i$. $\mathbf{W}_e$ represents the encoder's weights, and $g_e(\cdot)$ denotes the encoding function. Given the input sample $\mathbf{x}'_i$, the compressed representation $\mathbf{f}_i$ is as follows:

$$\mathbf{f}_i = g_e(\mathbf{x}'_i; \mathbf{W}_e). \quad (16)$$

- **Decoder**: The decoder maps the compressed representation $\mathbf{f}_i$ from the feature space $\varphi(\mathbf{x}')$ back to the original sample space X^M , reconstructing the full representation. The decoding process is defined as $g_d(\cdot; \mathbf{W}_d) : \varphi(\mathbf{x}') \rightarrow X^M$, where $\mathbf{W}_d$ represents the decoder's weights, and $g_d(\cdot)$ denotes the decoding function. Given the compressed representation $\mathbf{f}_i$, the reconstructed sample $\tilde{\mathbf{x}}_i$ is obtained as follows:

$$\tilde{\mathbf{x}}_i = g_d(\mathbf{f}_i; \mathbf{W}_d). \quad (17)$$

Finally, we obtain the compressed representation $\mathbf{f}$ encoded by $\mathcal{A}_e$ and the reconstructed outputs $\tilde{\mathbf{x}}$, which conforms to the original data distribution, to construct the feature space $\varphi(\mathbf{x}')$ for the original samples.

2) *Contrastive Branch*: The reconstruction process of $\mathcal{A}_e$ facilitates the construction of global features for normal data; however, it lacks the ability to capture fine-grained features and establish local connections between normal samples. Therefore, we design the following steps to address these limitations.

First, $\mathbf{x}'_1$ and $\mathbf{x}'_2$ are encoded into latent compressed representations $\mathbf{f}_1$ and $\mathbf{f}_2$ by two separate encoders, denoted as

$Encoder_1$ and $Encoder_2$, respectively. The encoding process is represented as:

$$\begin{cases} \mathbf{f}_1 = g_{e1}(\mathbf{x}'_1; \mathbf{W}_{e1}), \\ \mathbf{f}_2 = g_{e2}(\mathbf{x}'_2; \mathbf{W}_{e2}), \end{cases} \quad (18)$$

where, g_{e1} and g_{e2} represent the encoding processes, with $\mathbf{W}_{e1}$ and $\mathbf{W}_{e2}$ denoting the weights of the $Encoder_1$ and $Encoder_2$, respectively:

Next, the compressed representation $\mathbf{f}_1$ and $\mathbf{f}_2$ are further compressed through an additional encoding step, resulting in $\mathbf{q}_1$ and $\mathbf{q}_2$:

$$\begin{cases} \mathbf{q}_1 = H_1(\mathbf{f}_1; \mathbf{W}_{h1}), \\ \mathbf{q}_2 = H_2(\mathbf{f}_2; \mathbf{W}_{h2}). \end{cases} \quad (19)$$

Here, H_1 and H_2 represent the compression encoding processes, and $\mathbf{W}_{h1}$ and $\mathbf{W}_{h2}$ denote the weights of the corresponding compression encoding layers. The compressed features $\mathbf{q}_1$ and $\mathbf{q}_2$ are aligned with the dimensions of the original data classes and are used as positive encoded pairs in the subsequent feature self-contrastive strategy.

Specifically, the feature self-contrastive strategy computes the similarity between positive encoding pairs $\mathbf{q}_1^i$ and $\mathbf{q}_2^i$, and negative samples stored in a memory queue $\mathbf{q}_2^t$ to $\mathbf{q}_2^k$ ($i \in [t \sim k]$). The similarity between $\mathbf{q}_1^i$ and $\mathbf{q}_2^t$ to $\mathbf{q}_2^k$ is minimized, encouraging the $\mathcal{A}_e$ to differentiate features in normal data and establish specific correlations. Simultaneously, the similarity between $\mathbf{q}_1^i$ and $\mathbf{q}_2^i$ is maximized to ensure consistency of features for the same data across different augmentation modes. It is noteworthy that at the beginning of training, we initialize a memory queue to store $\mathbf{q}_2^i$. Additionally, by imposing constraints on the deep encoded features $\mathbf{q}_1$ and $\mathbf{q}_2$, the strategy enhances the representational capacity of the latent features $\mathbf{f}_1$ and $\mathbf{f}_2$, thus improving the discriminative ability between positive and negative samples. The process of the feature self-contrastive strategy is concisely presented in Algorithm 1.

Overall, this dual-branch optimization strategy not only differentiates between positive and negative samples, but also establishes constraint relationships among positive samples. This approach helps in identifying anomalous data by leveraging the advantages of unsupervised learning in distinguishing unknown data. At the same time, it mitigates the overfitting issue of labeled samples in supervised learning, thereby improving the model's generalization performance on unseen data.

Finally, we extract several feature factors: the enhanced compressed representation $\mathbf{f}_1$, the reconstruction error e , computed as the L2 norm of the difference between the input and its reconstruction ($e = \|\mathbf{x}' - \tilde{\mathbf{x}}\|_2$), and the reconstruction residual vector ϵ , which characterizes the directional component of the reconstruction error. These factors are combined to form the feature set $\mathbf{x}_{cat} = [\mathbf{f}_1, e, \epsilon]$. This set $\mathbf{x}_{cat}$ encapsulates the essential representations of the data, thereby guiding the model to effectively identify anomalies.

D. Anomaly Probability Generation Module

We integrate the APM and SCLM into a unified framework, where the parameters of $\mathcal{A}_e$ are initialized with pre-trained

Algorithm 1 Dual-Branch Training for Self-Contrastive Enhancement Learning

Parallel branch 1: Reconstruction
Inputs: Augmented data $\mathbf{x}'_1 \in \mathbb{R}^{M \times batch}$
Outputs: Compressed representations $\mathbf{f}_1 \in \mathbb{R}^m$, reconstruction outputs $\tilde{\mathbf{x}}_1$, and reconstruction loss L_{RE}

- 1: Randomly initialize query encoder g_{e1} and query decoder g_{d1}
- 2: **for** $i = 1$ to $epoch_{1st}$ **do**
- 3: **for** each minibatch $\mathbf{x}'_1$ **do**
- 4: Encode input to compressed representations $\mathbf{f}_1$
- 5: Decode $\mathbf{f}_1$ to reconstruction outputs $\tilde{\mathbf{x}}$
- 6: Compute reconstruction loss L_{RE} between $\mathbf{x}'_1$ and $\tilde{\mathbf{x}}$
- 7: Update encoder g_{e1} and decoder g_{d1} by backpropagation w.r.t. L_{RE}
- 8: **end for**
- 9: **end for**
- 10: **Return:** Compressed representations $\mathbf{f}_1$, reconstruction outputs $\tilde{\mathbf{x}}$, and reconstruction loss L_{RE}

Parallel branch 2: Contrastive
Inputs: Augmented data $\mathbf{x}'_1 \in \mathbb{R}^{M \times batch}$, $\mathbf{x}'_2 \in \mathbb{R}^{M \times batch}$, query encoder g_{e1} , key encoder g_{e2} , memory queue Q
Outputs: Compressed representations $\mathbf{f}_1 \in \mathbb{R}^m$

- 1: Randomly initialize g_{e1} , g_{e2} , and memory queue Q
 - 2: **for** $i = 1$ to $epoch_{1st}$ **do**
 - 3: **for** each minibatch $(\mathbf{x}'_1, \mathbf{x}'_2)$ **do**
 - 4: Encode input to compressed representations $\mathbf{f}_1, \mathbf{f}_2$
 - 5: Encoding compressed features:

$$\mathbf{q}_1 \leftarrow H_1(\mathbf{f}_1), \quad \mathbf{q}_2 \leftarrow H_2(\mathbf{f}_2)$$
 - 6: Compute positive logit: $l^+ \leftarrow \mathbf{q}_1 \cdot \mathbf{q}_2$
 - 7: Compute negative logits: $l^- \leftarrow \mathbf{q}_1 \cdot \mathbf{Q}$
 - 8: Compute InfoNCE loss L_{NCE} from l^+ and l^-
 - 9: Update query encoder g_{e1} by backpropagation w.r.t. L_{NCE}
 - 10: Momentum update key encoder g_{e2} from g_{e1}
 - 11: Enqueue new key $\mathbf{q}_2$ into Q
 - 12: Dequeue the oldest key from Q to maintain its size at K
 - 13: **end for**
 - 14: **end for**
 - 15: **Return:** Compressed representations $\mathbf{f}_1$
-

weights. The APM consists of a multi-layer perceptron (MLP), which transforms the feature set $\mathbf{x}_{cat}$ into an anomaly probability y' . Specifically, the feature set $\mathbf{x}_{cat}$ is fed into the first layer of the MLP, $\mathbf{z}_1$, to learn the weights between different feature components. We adopt a feature-enhanced MLP architecture to amplify the focus on low-dimensional feature e . Specifically, the low-dimensional feature e is integrated into each layer of the MLP, where the structure of each layer $\mathbf{z}_j$ is described as follows:

$$\mathbf{z}_j = \text{MLP}(\mathbf{w}_z^j \mathbf{z}_{j-1} + \mathbf{w}_e^j e + \mathbf{b}_j), \quad (20)$$

where j is the layer index, $\mathbf{w}_z^j$ and $\mathbf{w}_e^j$ represent the weights of each layer, and $\mathbf{b}_j$ is the bias, and $\mathbf{z}_1 = (\mathbf{w}_f^1 \mathbf{f}_1 + \mathbf{w}_\epsilon^1 \epsilon + \mathbf{w}_e^1 e + \mathbf{b}_1)$, where $\mathbf{w}_f^1$ and $\mathbf{w}_\epsilon^1$ are the weights as well. This enhanced

structure effectively balances the weights among different factors, enabling the model to achieve better performance in more complex tasks.

After a series of dimensionality reduction steps, the output layer $\mathbf{z}_{out}$ generates an anomaly probability y' . A value of y' close to 1 indicates a potential anomaly, while a value close to 0 suggests that the sample is normal.

E. Training Procedure

The training process for SEAD-Net is divided into two distinct stages, aiming to progressively optimize the model in identifying anomalous elements. In the first training stage, the SEAD-Net is pre-trained to simultaneously optimize two branches, acquiring enhanced latent representations of the data. In the second training stage, the SEAD-Net is trained in an end-to-end manner to accurately calculate the anomaly probability of the data.

1) *The First Training Stage:* In the reconstruction branch, the encoding-decoding process of $\mathcal{A}_e$ is optimized by the reconstruction error loss function L_{RE} :

$$L_{RE} = \|\tilde{\mathbf{x}}_1 - \mathbf{x}'_1\|_2. \quad (21)$$

Here, $\|\tilde{\mathbf{x}}_1 - \mathbf{x}'_1\|_2$ represents the Euclidean distance between the input sample $\mathbf{x}'_1$ and the reconstructed sample $\tilde{\mathbf{x}}_1$. A smaller Euclidean distance indicates better reconstruction performance of $\mathcal{A}_e$.

In the contrastive branch, the similarity computation process is optimized using the InfoNCE loss L_{NCE} :

$$L_{NCE} = -\log \left(\frac{\exp(\mathbf{q}_1^i \cdot \mathbf{q}_2^i / \tau)}{\sum_{j=0}^K \exp(\mathbf{q}_1^i \cdot \mathbf{q}_2^j / \tau)} \right), \quad (22)$$

where τ is the temperature parameter, which controls the sharpness of the learning process. When $\mathbf{q}_1^i$ is similar to the positive sample $\mathbf{q}_2^i$ and dissimilar to all negative samples in the memory queue $\mathbf{q}_2^t \sim \mathbf{q}_2^k$, the loss is minimized, where $j \in [t, k]$. Notably, we employ an exponential moving average technique to smoothly update the parameters from $\mathcal{A}_e^1$ to $\mathcal{A}_e^2$, which aids in learning more robust global representations.

The overall task objective for the first stage of the dual-branch training is as follows: 1) Maximize the fitting accuracy in the positive sample feature space by minimizing the reconstruction error e when training $\mathcal{A}_e$; 2) Maximize the similarity between positive encoding pairs, i.e., maximize the similarity between $\mathbf{q}_1^i$ and $\mathbf{q}_2^i$; 3) Maximize the distinction between positive samples by minimizing the similarity between the positive sample encoding $\mathbf{q}_1^i$ and the negative sample encoding set $\mathbf{q}_2^t \sim \mathbf{q}_2^k$. Combining Eqs. (21) and (22), the total loss function for the first training stage, L_{1st} , can be obtained as follows:

$$\begin{aligned} L_{1st} &= \lambda \cdot L_{RE} + (1 - \lambda) \cdot L_{NCE} \\ &= \lambda \cdot \|\tilde{\mathbf{x}}_1 - \mathbf{x}'_1\|_2 - (1 - \lambda) \cdot \log \left(\frac{\exp(\mathbf{q}_1^i \cdot \mathbf{q}_2^i / \tau)}{\sum_{j=0}^K \exp(\mathbf{q}_1^i \cdot \mathbf{q}_2^j / \tau)} \right), \end{aligned} \quad (23)$$

where the hyperparameter λ balances the optimization weights between the two tasks. This strategy ensures that the training

TABLE I
DETAILED INFORMATION ABOUT VARIOUS DATASETS

Datasets	Num	Dim	Train	Test	Ano
NSL-KDD	148,517	122	118814	29703	0.05%
Shuttle	49,097	9	39278	9819	0.08%
Fraud	284,807	29	227846	56961	0.01%
Bank	41,188	20	32950	8238	0.10%
Census	299,285	500	239428	59857	0.01%
CelebA	202,599	39	162079	40520	0.02%
Thyroid	7,200	21	5760	1440	0.55%
Mammography	11,183	6	8946	2237	0.34%

process can be tailored to each dataset, thereby enhancing the model's effectiveness and applicability.

2) *The Second Training Stage*: In the second stage of training, we optimize the complete SEAD-Net using a conditional loss function, with the objective as follows: 1) Cluster the normal samples around a region centered at zero; 2) Push the anomalous samples away from this central region. The second-stage training loss L_{2nd} is defined as:

$$L_{2nd} = \sum_i (1 - y_i)(|\mathcal{F}(\mathbf{x}_{cat})| + e_i) + y_i(\max(0, a_0 - \mathcal{F}(\mathbf{x}_{cat})) + \max(0, a_0 - e_i)). \quad (24)$$

Here, $\mathcal{F}(\mathbf{x}_{cat})$ represents the output of the APM, and y_i denotes the label of the data. It can be seen that the optimization process effectively constrains the reconstruction error and anomaly probability. When a sample is normal, i.e., $y_i = 0$, the model optimizes the first part of the loss; when a sample is anomalous, i.e., $y_i = 1$, the model optimizes the second part. a_0 is a predefined constant that controls the deviation of anomalous samples, preventing the model from over-optimizing for anomalies. In essence, the first part of the loss restricts the latent features and anomaly probabilities of normal samples within a hypersphere centered at zero, while the second part ensures that anomalous samples significantly deviate from the distribution of normal data, maintaining a reasonable distance.

In summary, the complete two-stage training procedure of SEAD-Net is provided in Section S-I of the **Supplementary Material**.

IV. EXPERIMENTS

This section first introduces the datasets and experimental settings. Subsequently, we discuss the performance of SEAD-Net.

A. Dataset

We evaluate the performance of the proposed SEAD-Net on multiple public datasets, which cover applications in network security identification, financial service fraud detection, and bank marketing detection. Detailed datasets' information can be found in Tab. I, where *Ano* denotes the percentage of anomalous samples in the training data. Further details on the individual datasets are provided in Section S-II-A of the **Supplementary Material**. Across all datasets, anomalous

samples are defined according to fixed anomaly criteria or substantial deviations from the majority class, thereby ensuring that the anomaly labels remain consistent with dataset-specific semantics.

To improve cross-dataset comparability and training stability, we apply a unified preprocessing pipeline to all datasets, including mean imputation for missing values, one-hot encoding for categorical attributes, and feature normalization.

B. Experimental Setup

We evaluate the performance of the proposed SEAD-Net against several state-of-the-art anomaly detection algorithms, including Copod [26], Ecod [27], iForest [28], DIForest [29], REPEN⁺ [39], FSNet⁺ [38], DevNet [40], FEWAE [41], Dual-MGAN [43], and Deep SAD [42]. Consistent with FEWAE, we use 80% of the samples for training and 20% for testing, while maintaining the same anomaly ratio in both sets, and perform weakly supervised training with 30 anomalous samples. We adopt AUC-ROC and AUC-PR as evaluation metrics, following the protocol in [45], and additionally report the F1-score to further assess model reliability. For reproducibility, models are saved using the optimal parameter settings.

Detailed hyperparameter settings and implementation details are provided in Section S-II-B of the **Supplementary Material**. Moreover, the **Supplementary Material** and source code are available at: <https://github.com/pumpkinyouge/SEAD-Net>.

C. Comparison Results

Tab. II shows the comparison of SEAD-Net with various competitive methods in terms of AUC-PR and AUC-ROC metrics. It is worth noting that FSNet⁺ and REPEN⁺ are manually tuned versions of FSNet and REPEN, respectively. The last row shows the average performance of different competitive methods across all datasets.

According to Tab. II, we can see that SEAD-Net demonstrates significant performance improvements compared to all competing methods. Specifically, for AUC-PR, SEAD-Net surpasses the competing methods by 34.35% (Copod), 34.23% (Ecod), 38.13% (iForest), 44.21% (DIForest), 24.34% (REPEN), 19.81% (Dual-MGAN), 37.25% (FSNet), 16.47% (Deep-SAD), 16.76% (DevNet), and 16.28% (FEWAE), on average. Regarding AUC-ROC performance, SEAD-Net also shows remarkable advantages, outperforming the competing methods by 17.36% (Copod), 16.42% (Ecod), 21.92% (iForest), 26.49% (DIForest), 10.97% (REPEN), 8.85% (Dual-MGAN), 18.50% (FSNet), 7.54% (Deep-SAD), 5.95% (DevNet), and 5.96% (FEWAE), respectively.

Furthermore, we also compare the best results of all comparative algorithms on each dataset. In terms of AUC-PR values, SEAD-Net demonstrates advantages over the best results, with improvements of 1.26% (NSL-KDD), 1.57% (Shuttle), 7.49% (Fraud), 12.50% (Bank), 6.19% (Census), 4.96% (CelebA), 45.92% (Thyroid), and 17.26% (Mammography). For AUC-ROC values, SEAD-Net shows advantages over the best results, with improvements of 1.18% (NSL-KDD), 0.09% (Shuttle), 1.16% (Fraud), 6.84% (Bank), 5.55% (Census), 0.66% (CelebA), 14.45% (Thyroid), and 1.33% (Mammography). It is noteworthy that on datasets where AUC-PR values

TABLE II

COMPARISON OF THE AUC-PR AND AUC-ROC PERFORMANCE OF SEAD-NET WITH VARIOUS COMPETING METHODS. THE BEST RESULT IS HIGHLIGHTED BY BOLDFACE AND UNDERLINING, WHILE THE SECOND-BEST RESULT IS UNDERLINED

AUC-PR											
Datasets	Copod [26]	Ecod [27]	iForest [28]	DIForest [29]	REPEN ⁺ [39]	Dual-MGAN [43]	FSNet ⁺ [38]	Deep-SAD [42]	DevNet [40]	FEWAE [41]	SEAD-Net (Our approach)
NSL-KDD	0.5131	0.5212	0.6600	0.5324	0.9560	0.9530	0.9658	0.9685	0.9471	<u>0.9715</u>	0.9841 (1.26%↑)
Shuttle	0.9509	0.8765	0.9049	0.5933	0.9593	0.9725	0.9670	0.9529	0.9585	<u>0.9745</u>	0.9902 (1.57%↑)
Fraud	0.3543	0.3259	0.2532	0.4088	0.6730	<u>0.7230</u>	0.0431	0.5776	0.6897	0.6916	0.7979 (7.49%↑)
Bank	0.4175	0.4016	0.3276	0.3555	0.3306	0.3928	0.1934	<u>0.4569</u>	0.3814	0.4284	0.5819 (12.50%↑)
Census	0.0728	0.0707	0.0766	0.0499	0.1649	0.2387	0.1927	0.3117	<u>0.3202</u>	0.2840	0.3821 (6.19%↑)
CelebA	0.1566	0.1509	0.0642	0.0698	0.1613	0.2147	0.0854	0.1543	<u>0.2795</u>	0.2278	0.3291 (4.96%↑)
Thyroid	0.1152	0.2299	0.1653	0.0288	0.0940	0.1653	0.1169	<u>0.4936</u>	0.2739	0.3135	0.9528 (45.92%↑)
Mammography	0.4597	0.4730	0.2865	0.2128	0.5023	0.5434	0.2438	0.5552	<u>0.5977</u>	0.5947	0.7703 (17.26%↑)
Average	0.3800	0.3812	0.3423	0.2814	0.4802	0.5254	0.3510	0.5588	0.5560	<u>0.5608</u>	0.7236 (16.28%↑)
AUC-ROC											
Datasets	Copod [26]	Ecod [27]	iForest [28]	DIForest [29]	REPEN ⁺ [39]	Dual-MGAN [43]	FSNet ⁺ [38]	Deep-SAD [42]	DevNet [40]	FEWAE [41]	SEAD-Net (Our approach)
NSL-KDD	0.5527	0.5809	0.7020	0.4552	0.9684	0.9612	0.9504	<u>0.9714</u>	0.9560	0.9540	0.9832 (1.18%↑)
Shuttle	<u>0.9966</u>	0.9942	0.9399	0.9726	0.9653	0.9744	0.9809	0.9825	0.9769	0.9831	0.9975 (0.09%↑)
Fraud	0.9443	0.9456	0.9524	0.9625	0.9729	0.9662	0.7347	0.9472	0.9796	<u>0.9821</u>	0.9937 (1.16%↑)
Bank	0.7971	0.7954	0.7314	0.7291	0.7234	0.7835	0.6231	0.8330	0.8073	<u>0.8573</u>	0.9257 (6.84%↑)
Census	0.6720	0.6577	0.6238	0.5871	0.7944	<u>0.8358</u>	0.7314	0.8001	0.8287	0.8275	0.8913 (5.55%↑)
CelebA	0.7504	0.7567	0.6971	0.6570	0.8940	0.8741	0.8076	0.7955	<u>0.9508</u>	0.9370	0.9574 (0.66%↑)
Thyroid	0.6933	0.7480	0.6883	0.4905	0.5792	0.6731	0.5637	<u>0.8523</u>	0.7835	0.7446	0.9968 (14.45%↑)
Mammography	0.9268	0.9300	0.6338	0.7493	0.9467	0.9456	0.8500	0.9368	<u>0.9632</u>	0.9599	0.9765 (1.33%↑)
Average	0.7917	0.8011	0.7461	0.7004	0.8555	0.8767	0.7802	0.8898	<u>0.9058</u>	0.9057	0.9653 (5.95%↑)

are generally low such as Fraud, Bank, Census, CelebA, Thyroid, and Mammography, SEAD-Net consistently achieves improvements by double digits compared to the best results. On NSL-KDD and Shuttle datasets, SEAD-Net also significantly enhances performance from high AUC-PR values of 0.9715 and 0.9745 to 0.9841 and 0.9902 respectively.

Furthermore, we present F1-score comparisons between our approach and competing methods on the Bank, CelebA, Fraud, and Mammography datasets in Section S-III of the **Supplementary Material**. Under the same experimental setup, these results consistently validate that SEAD-Net achieves a clear and consistent overall advantage over representative baselines.

D. Computational Complexity

To evaluate the practical applicability and deployment feasibility of SEAD-Net, we conduct a comprehensive computational complexity analysis by comparing it with five representative baseline models across multiple dimensions, including floating-point operations (FLOPs), parameter count, per-sample training time, per-sample inference time, and GPU memory consumption. For a fair comparison, the computational costs of two-phase methods are aggregated. The quantitative results are summarized in Tab. III.

As shown in Tab. III, SEAD-Net exhibits higher spatial complexity, measured in terms of parameter count and memory usage, than most baselines, which is primarily attributed to its dual-branch architecture. Despite this increase, the peak GPU memory consumption of SEAD-Net remains within the lightweight regime (below 190 MB), ensuring compatibility with common resource-constrained deployment scenarios. In addition, SEAD-Net demonstrates outstanding temporal efficiency: it achieves the fastest per-sample training time among all evaluated two-phase methods and maintains

TABLE III

COMPARISON OF THE COMPUTATIONAL COMPLEXITY OF SEAD-NET AND MATCHING ALGORITHMS

Computational Complexity							
Datasets		NSL-KDD	Shuttle	Fraud	Bank	Census	CelebA
Dim		122	9	29	20	500	39
REPEN ⁺	FLOPs/M	0.10	0.04	0.05	0.04	0.29	0.05
	Params/k	96.11	37.80	48.12	43.47	291.15	53.28
	Train_T/us	413.54	354.35	309.50	383.04	278.06	313.33
	Inference_T/us	18.32	6.18	5.57	8.59	6.53	14.37
	Peak_GPU/MB	89.27	18.69	49.87	20.34	597.64	47.84
FSNet ⁺	FLOPs/M	0.01	0.01	0.01	0.01	0.02	0.01
	Params/k	8.46	4.84	5.48	5.19	20.55	5.80
	Train_T/us	53.21	65.03	91.87	20.10	101.92	64.17
	Inference_T/us	61.51	83.12	55.37	24.85	54.69	72.61
	Peak_GPU/MB	8.24	8.20	8.20	8.20	8.53	8.21
DevNet	FLOPs/M	0.12	0.01	0.03	0.02	0.50	0.04
	Params/k	124.00	11.00	31.00	22.00	502.00	41.00
	Train_T/us	153.82	244.42	127.42	117.80	227.35	101.49
	Inference_T/us	13.77	17.37	17.93	11.21	15.79	16.05
	Peak_GPU/MB	17.92	16.40	16.63	16.52	22.62	16.79
Deep-SAD	FLOPs/M	0.15	0.09	0.10	0.09	0.34	0.10
	Params/k	144.38	86.53	96.77	92.16	337.92	101.89
	Train_T/us	104.45	99.98	79.83	105.83	102.85	100.07
	Inference_T/us	0.02	0.02	0.01	0.02	0.02	0.01
	Peak_GPU/MB	37.20	35.21	35.57	35.40	43.85	35.74
FEWAE	FLOPs/M	0.06	0.02	0.02	0.02	0.39	0.03
	Params/k	63.06	19.11	25.03	22.26	395.70	28.29
	Train_T/us	149.48	135.24	145.08	145.54	126.76	132.90
	Inference_T/us	2.18	2.95	1.92	3.56	2.15	1.92
	Peak_GPU/MB	36.85	33.75	34.25	34.03	52.91	34.51
SEAD-Net	FLOPs/M	0.52	0.30	0.34	0.32	1.24	0.36
	Params/k	521.18	303.88	342.34	325.03	1248.07	361.57
	Train_T/us	39.54	42.64	45.81	42.89	58.75	47.41
	Inference_T/us	0.97	0.95	0.86	1.01	1.10	0.89
	Peak_GPU/MB	172.66	167.03	168.03	167.58	189.92	168.06

microsecond-level inference latency, significantly outperforming DevNet and FEWAE.

V. DISCUSSIONS

A. Sample Efficiency

To evaluate the sample efficiency of SEAD-Net, we vary the number of labeled anomaly samples in the training set

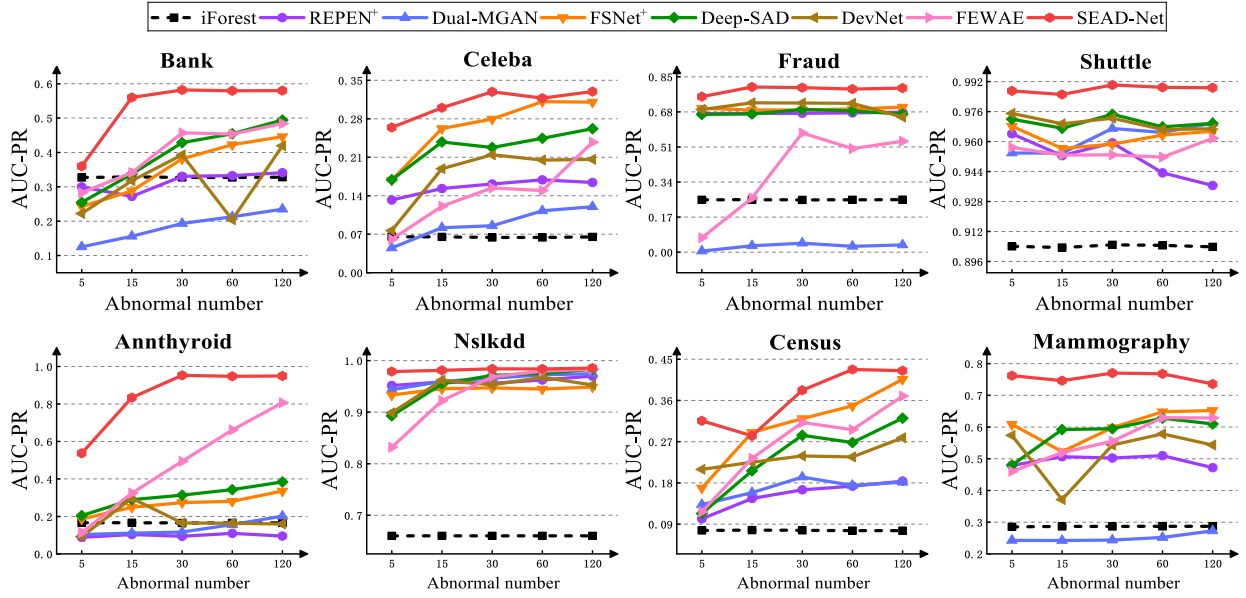


Fig. 3. Comparison of the AUC-PR performance of SEAD-Net and competing methods with different numbers of labeled anomaly samples. As shown in the figure, SEAD-Net (red solid line with markers) achieves high-precision anomaly detection with extremely low labeling costs. It consistently outperforms other methods across all labeling levels, demonstrating its superior sample efficiency.

from 5 to 120, while keeping all other experimental settings unchanged. The AUC-PR performance under different labeling budgets is reported in Fig. 3.

As shown in Fig. 3, SEAD-Net consistently outperforms competing methods across all labeling regimes, particularly under limited anomaly supervision, demonstrating strong robustness and effective utilization of scarce anomaly samples. While most methods achieve their peak performance when sufficient anomaly labels are available, their performance degrades noticeably as the number of labeled anomalies decreases. In contrast, SEAD-Net maintains stable and superior detection capability even under low-label conditions.

More importantly, except for the Census dataset, SEAD-Net trained with only 30 labeled anomaly samples achieves higher AUC-PR performance than all competing methods trained with 120 labeled anomalies, with improvements of 0.06% (NSL-KDD), 2.04% (Shuttle), 9.48% (Fraud), 8.76% (Bank), 1.89% (CelebA), 14.63% (Thyroid), and 11.81% (Mammography). When only 15 labeled anomalies are available on the Census dataset, SEAD-Net exhibits a notable performance drop, which may be attributed to noise introduced by newly labeled anomalies or substantial distributional divergence among anomaly samples, thereby hindering effective distribution fitting.

Overall, these results indicate that SEAD-Net can more effectively exploit a limited number of anomaly samples, leading to improved representation learning, stronger generalization performance, and enhanced robustness compared to existing methods.

B. Ablation Study

1) *Comparison of the Performance of Different SEAD-Net Variants:* We introduce the variant V_{SCLM} , which contains only the SCLM module, to examine the impact of removing the customized DEM from SEAD-Net. To further assess

TABLE IV
COMPARISON OF THE AUC-ROC AND AUC-PR PERFORMANCE OF SEAD-Net AND ITS VARIANTS WITH VARIOUS COMPETING METHODS

AUC-ROC						
Datasets	NSL-KDD	Shuttle	Bank	Census	CelebA	Thyroid
DevNet	0.9560	0.9769	0.8073	0.8287	0.9508	0.7835
FEWAE	0.9540	0.9831	0.8573	0.8275	0.9370	0.7446
V_{contra}	0.9799	0.9916	0.9238	0.8722	0.9565	0.9958
V_{recon}	0.9810	0.9971	0.9228	0.8854	0.9582	0.9949
V_{recon_D}	0.9810	0.9971	0.9242	0.8897	0.9563	0.9951
V_{SCLM}	0.9811	0.9975	0.9236	0.8885	0.9594	0.9939
SEAD-Net	0.9832	0.9975	0.9257	0.8913	0.9574	0.9968

AUC-PR						
Datasets	NSL-KDD	Shuttle	Bank	Census	CelebA	Thyroid
DevNet	0.9471	0.9585	0.3814	0.3202	0.2795	0.2739
FEWAE	0.9715	0.9745	0.4284	0.2840	0.2278	0.3135
V_{contra}	0.9805	0.9829	0.5762	0.3510	0.3002	0.9455
V_{recon}	0.9832	0.9897	0.5705	0.3633	0.3065	0.9490
V_{recon_D}	0.9827	0.9897	0.5792	0.3753	0.3164	0.9492
V_{SCLM}	0.9843	0.9902	0.5737	0.3739	0.3101	0.9372
SEAD-Net	0.9841	0.9902	0.5819	0.3821	0.3291	0.9528

the joint reconstruction–contrastive dual-branch design within SCLM, we construct additional variants by removing individual branches: V_{recon} , which retains only the reconstruction branch, and V_{contra} , which retains only the contrastive branch. In addition, by incorporating the customized DEM into V_{recon} , we obtain V_{recon_D} to evaluate the effectiveness of DEM under a baseline reconstruction setting. All variants are compared with the full SEAD-Net, DevNet, and FEWAE under consistent experimental settings.

As reported in Tab. IV, V_{recon_D} outperforms V_{recon} , achieving an average AUC-PR improvement of 0.51%, indicating that DEM enhances representation learning by introducing data diversity. Similarly, V_{SCLM} outperforms V_{recon} on most datasets, with an average AUC-PR improvement of 0.12%, validating the advantage of the dual-branch structure over

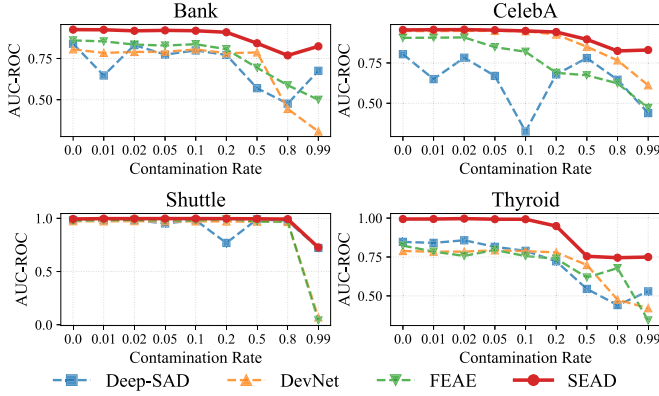


Fig. 4. Comparison of the AUC-ROC performance of SEAD-Net with baseline methods under varying anomaly contamination rates ($c \in [0.01, 0.99]$).

single-branch reconstruction. Notably, SEAD-Net achieves the largest overall gain, with an average AUC-PR improvement of 0.97%, exceeding the combined improvements of $V_{\text{recon_D}}$ and V_{SCLM} , which indicates a strong synergistic effect between the customized DEM and self-contrastive learning.

Moreover, across all datasets, V_{recon} consistently outperforms V_{contra} , suggesting that the reconstruction branch serves as the primary driver of performance, while the contrastive branch functions as an effective regularizer.

2) *Robustness Analysis Against Anomaly Contamination:* We evaluate the robustness of SEAD-Net under label noise using a controlled anomaly contamination setting. Synthetic anomalies are generated via a Feature Swapping strategy with a small swap ratio of 0.05 and injected into the training set as normal samples, while the contamination rate varies from 0.01 to 0.99. Fig. 4 compares the AUC-ROC performance of SEAD-Net with three baselines (Deep-SAD, DevNet, and FEWAE) under increasing anomaly contamination.

As the contamination rate increases, the performance of all methods degrades. Deep-SAD exhibits pronounced instability, while DevNet and FEWAE deteriorate rapidly at moderate contamination levels. In contrast, SEAD-Net shows a smoother degradation trend and consistently maintains a clear performance margin, even under extreme contamination. These results suggest that the self-contrastive learning mechanism effectively regularizes the latent space, thereby enabling robust anomaly detection in the presence of severe label noise.

3) *Visualization of Dataset-Specific Enhancement Effects:* To analyze the effectiveness of the proposed enhancement library, we define the Performance Gain (PG) as the performance difference between each enhancement setting and the corresponding no-enhancement baseline. We then visualize the PG values of individual enhancement methods across five datasets to reduce heuristic dataset-specific augmentation selection. The resulting heatmap uses warm and cool colors to indicate positive and negative effects, respectively, with datasets and enhancement methods arranged along the horizontal and vertical axes in Fig. 5.

As shown in Fig. 5, individual enhancements exhibit dataset-dependent effectiveness and do not consistently benefit all datasets. Nevertheless, each enhancement provides advantages on a subset of datasets, and every dataset benefits from

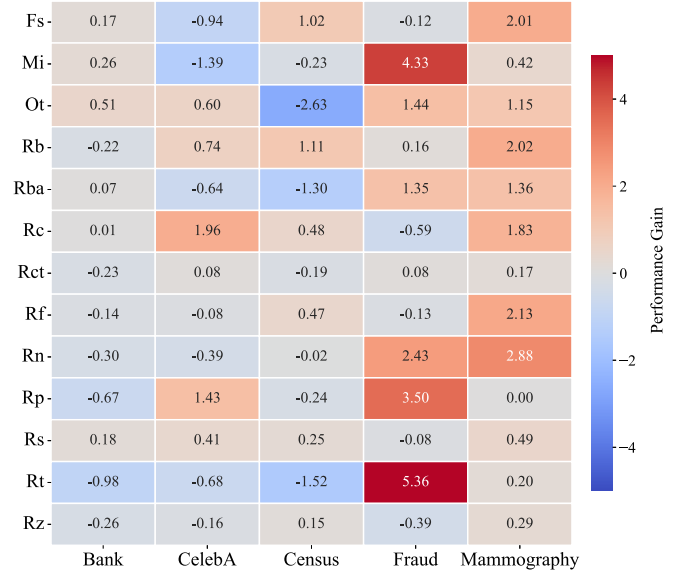


Fig. 5. Visualization of dataset-wise Performance Gain (PG) induced by individual enhancement methods.

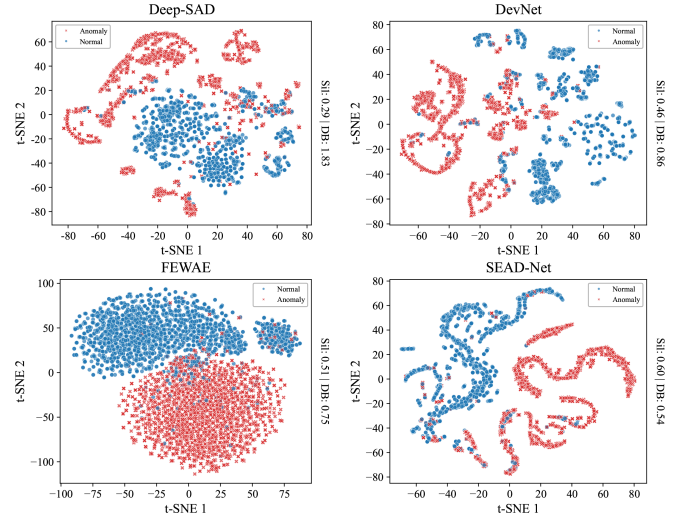


Fig. 6. Visualization of t-SNE projections of latent representations produced by SEAD-Net and baseline methods.

multiple enhancement choices. These observations indicate that, while individual augmentations have limited generality, the proposed enhancement library provides broad coverage and supports dataset-specific augmentation selection.

4) *t-SNE Visualization of Latent Representations:* To further examine whether the final SEAD-Net preserves representational advantages over competing methods, we visualize the learned latent representations using t-SNE. Unlike the previous visualization, which analyzes internal SEAD-Net variants, Fig. 6 compares the final SEAD-Net with three baselines, namely Deep-SAD, DevNet, and FEWAE.

As illustrated in Fig. 6, the latent representations produced by Deep-SAD and DevNet exhibit poorly defined cluster structures. FEWAE forms comparatively clearer class patterns; however, samples within the same class remain more dispersed. In contrast, SEAD-Net generates noticeably more compact intra-class representations while maintaining clear inter-class separation. This observation is further supported by

TABLE V
COMPARISON OF SEAD-NET PERFORMANCE UNDER DIFFERENT
ENHANCEMENT MODULES

	Method	NSL-KDD	Shuttle	Fraud	Bank	Census	CelebA	average
AUC-ROC	RODEO	0.9487	0.9922	0.9807	0.8131	0.8116	0.8835	0.9050
	TMG-GAN	0.9578	0.9848	0.9801	0.8965	0.8515	0.9517	0.9371
	SEAD-Net	0.9832	0.9975	0.9937	0.9257	0.8913	0.9574	0.9581
AUC-PR	RODEO	0.9350	0.9806	0.7251	0.4043	0.1799	0.1563	0.5635
	TMG-GAN	0.9607	0.9768	0.7129	0.5031	0.2675	0.2907	0.6186
	SEAD-Net	0.9841	0.9902	0.7979	0.5819	0.3821	0.3291	0.6776

quantitative clustering metrics: SEAD-Net achieves the highest Silhouette Coefficient (Sil) and the lowest Davies–Bouldin Index (DB), confirming superior intra-class compactness and inter-class separation. Overall, these results indicate that SEAD-Net preserves class separability while substantially enhancing intra-class compactness, leading to more clearly structured anomaly boundaries.

5) *Comparison of DEM With State-of-the-Art Enhancement Methods:* This subsection evaluates the improved DEM module in SEAD-Net. Under identical experimental settings, we compare it with two state-of-the-art enhancement approaches: a GAN-based data augmentation method (TMG-GAN) [37] and a diffusion-based data generation method (RODEO) [44]. The quantitative results are summarized in Tab. V.

From Tab. V, we observe that replacing DEM under the same settings leads to a significant performance degradation for SEAD-Net. On average, SEAD-Net outperforms diffusion-based models by 11.40% in AUC-PR and 5.32% in AUC-ROC, and surpasses GAN-based models by 5.89% in AUC-PR and 2.11% in AUC-ROC. This performance gap may stem from the limited ability of diffusion-based models to capture discriminative features in streaming data, as they are primarily designed for image generation. For GAN-based approaches, newly generated samples under extremely low-label conditions (e.g., 30 labeled samples) tend to distort the original feature distribution, thereby degrading detection performance.

Notably, both diffusion- and GAN-based methods exhibit dataset-specific strengths: diffusion models perform better on Shuttle and Thyroid, whereas GAN-based methods achieve stronger results on NSL-KDD, Bank, Census, and CelebA. Overall, the DEM module in SEAD-Net adopts customized multi-dimensional stochastic augmentation strategies, including random cropping, flipping, shifting, and mirroring. These strategies generate diverse, robust, yet faithful data variants while preserving the original feature distribution, even under extremely limited supervision.

Moreover, additional ablation evidence is provided in the **Supplementary Material**. Section S-IV-A shows that SEAD-Net remains stable across a broad range of a_0 values, with $a_0 = 1.5$ serving as an effective default choice, while Section S-IV-B indicates that better performance is achieved when $\lambda \in [0, 10^{-4}]$, highlighting the need to maintain a suitable balance between reconstruction and contrastive optimization. Section S-V-A further shows that appropriate pairwise augmentation combinations can yield additional gains, although overly strong transformations may introduce feature distortion on specific datasets. Finally, Section S-V-B presents the latent feature-space visualization of SEAD-Net variants, confirming

that the progressive introduction of customized enhancement and self-contrastive learning consistently improves representation quality and leads to clearer separation between normal and anomalous samples.

VI. CONCLUSION

This paper proposes SEAD-Net, a weakly supervised anomaly detection framework based on self-contrastive learning, which enhances the feature representation capability of data in the model’s feature space, thereby improving the accuracy and robustness of anomaly detection. Specifically, we enhance the representation ability of each sample through our customized data augmentation module. Simultaneously, we design a self-contrastive learning module to boost the model’s ability to learn complex patterns within the data. Through these complementary steps, SEAD-Net extracts features that effectively represent critical information from the raw data, thereby addressing the issues of inaccurate or insufficient data features. Compared with other WADMs, our SEAD-Net can significantly improve anomaly detection performance and sample utilization efficiency. Experimental results demonstrate that SEAD-Net outperforms other competing methods in both performance and labeled sample utilization rate.

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